

Probablility and Statistics
Course Code: STAT-2311
Chapter: 8

Definition:

1. Hypothesis
 - A statement about the nature of populations.
Example: Students who eat breakfast will perform better on a stat exam than students who do not eat breakfast.
2. Null Hypothesis
 - The hypothesis which we are going to test for possible rejection under the assumption that it is true. Null hypothesis is denoted by H_0 ; $H_0 : \mu_1 = \mu_2$.
3. Alternative Hypothesis
 - Each of all possible hypothesis other than null hypothesis is called alternative hypothesis and is usually denoted by H_1 ; $H_0 : \mu_1 \neq \mu_2$.
4. Type- I error
 - The error of rejecting H_0 (accepting H_1) which is true is called error of first kind or Type-I error.
5. Type- II error
 - The error of accepting null hypothesis H_0 when it is false is called error of 2nd kind or H_1 . Type-II error is denoted by β .
6. Level of significance(α)
 - The probability of type-I error is denoted by α is called level of significance.
7. P-value
 - The p-value is the least possible value of α . For which we can reject the null hypothesis.

Commonly used test statistic:

1. The normal test (z-test)
2. The t test
3. Chi-square (χ^2) test
4. F-test

Application:

Chi-square (χ^2) test

- a) To test the significance of a specified population variance

- b) To test the goodness of fit of a distribution
- c) To test the independence of attributes
- d) To test the equality of several correlation coefficients
- e) To test the homogeneity of several tests
- f) To test the homogeneity of several population variance

F-test

- a) To test the equality of several population means
- b) To test the equality of two population variance
- c) To test the significance of regression
- d) To test the significance of multiple correlation coefficient

Math Problem:

Problem, Chapter-8: Two hundred Engineers were interviewed and classified according to their results and job satisfaction. The distribution of graduates by results and job satisfaction are given in the following contingency table:

Results	Job satisfaction	
	Yes	No
Excellent	20	70
Good	45	65

Test at 5% level of significance whether there is any relationship between results and job satisfaction.

(Note: At 5% level of significance tabulated value of Chi-square @ 1df =3.84)

Solution: We consider the following hypothesis

H0: There is no relationship between results and job satisfaction.

H1: There is a relationship between results and job satisfaction.

Results	Job satisfaction		Total
	Yes	No	
Excellent	20 (O ₁₁)	70(O ₁₂)	90
Good	45(O ₂₁)	65(O ₂₂)	110
Total	65	135	200

We know,

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Here, $E_{11} = \frac{90 \times 65}{200} = 29.25$, $E_{12} = \frac{90 \times 135}{200} = 60.75$, $E_{21} = \frac{110 \times 65}{200} = 35.75$,

$$E_{22} = \frac{110 \times 135}{200} = 74.25$$

Therefore, $\chi^2 = \frac{(20 - 29.25)^2}{29.25} + \frac{(70 - 60.75)^2}{60.75} + \frac{(45 - 35.75)^2}{35.75} + \frac{(65 - 74.25)^2}{74.25}$

$$= 2.93 + 1.41 + 3.75 + 1.15 = 9.24 \quad [\chi^2_{.05(1)} = 3.84]$$

Comment: Reject H₀, There is a relationship between results and job satisfaction.